

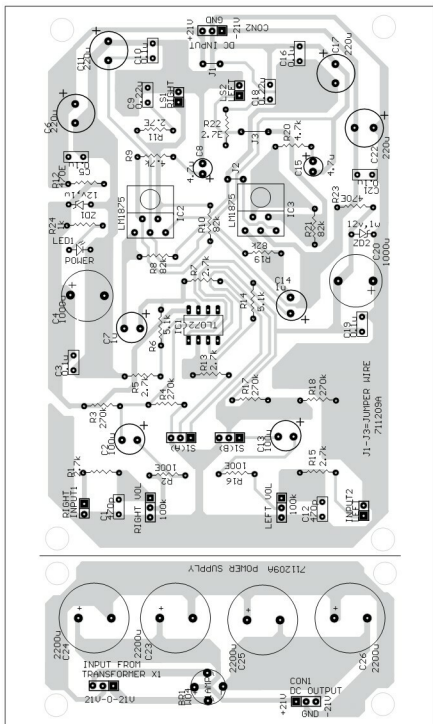


Fig. 5: Component layout for the PCB

Use the insulation sleeves, flat washers, lock washers and nuts, so that everything is in place before screws are tightened. Mount volume control knobs into their respective holes on the front panel. (You may need to slip a couple of thin washers into the spindles of the knobs,

to prevent them from protruding inside the cabinet.) With the PCB fitted at the bottom of the cabinet, bring speaker wires from connecting points, marked as LS1 and LS2, respectively, on the PCB, to the rear panel.

Fit four screws along with suitable

spacers into the holes located in four corners of the cabinet bottom to tighten up the PCB firmly in its place. You can place the power transformer inside the cabinet and tighten it to the cabinet bottom with nuts and bolts.

Now take a pair of tweezers or needle nose pliers and carefully push the tip of each wire through holes for left and right RCA input sockets and stereo/mono switch, respectively. Fix the sockets and switch firmly to the cabinet by tightening the respective nuts. Make sure that the wires are not twisted together or unduly strained. Also, make sure that sufficient length of wires from the PCB to the cabinet are provided for speakers, switches, volume controls and power supply. Following these steps, solder each wire to the PCB pad and clip off any excess parts. This essentially completes the wiring.

Testing brick amplifier

As there are no adjustments to be made, connect the amplifier to a pair of speakers, source of power and provide mono or stereo signals, before listening to the sound.

It is a good idea to perform a functional check of the power supply before IC TL072 (IC1) is inserted into its socket on the PCB. So, turn on the power, then measure the voltages across the supply, which should give +21V and -21V outputs, respectively. Now try checking the voltages at pins 8 and 4 of the socket for IC1, which should be +12V and -12V, respectively. If all the four voltages fulfill the measurement criteria, then turn off the power, unplug the mains cord, insert IC1 in the socket and fit the top of the cabinet.

If you are concerned about the proximity of the sleeved transformer's primary connections to the side of the cabinet, you may attach a protective inverted-L cover of 'elephant hide' or a similar fireproof insulating sheet over the side facing the transformer.

The brick amplifier is now ready for use. ●